Table 2: AUC scores for error type classification (TriviaQA). Error types are predictable from the inner model representations, indicating the encoding of fine-grained information on errors.

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	$0.86_{\pm 0.002}$	$0.85_{\pm 0.011}$	$0.87_{\pm 0.002}$	$0.88_{\pm 0.014}$
(B) Consistently correct	$0.88_{\pm 0.001}$	$0.82_{\pm 0.008}$	$0.86_{\pm 0.001}$	$0.81_{\pm 0.002}$
(C) Consistently incorrect	$0.59_{\pm 0.002}$	$0.67_{\pm 0.002}$	$0.59_{\pm 0.002}$	$0.64_{\pm 0.003}$
(D) Two competing	$0.63_{\pm 0.002}$	$0.68_{\pm 0.006}$	$0.61_{\pm 0.001}$	$0.65_{\pm 0.004}$
(E) Many answers	$0.90_{\pm 0.001}$	$0.84_{\pm 0.003}$	$0.89_{\pm 0.001}$	$0.89_{\pm 0.001}$

output correctness but also fine-grained information about potential errors. While detection performance varies between types, the predictability of each type is valuable on its own, as it opens the possibility of tailoring targeted interventions for specific error types. Additionally, although performance on error types C and D is lower, it remains well above random, providing meaningful insights. These results suggest that internal representations encode more than just binary correctness, revealing a nuanced taxonomy of error types and offering deeper insights into how these models process and encode knowledge.

6 Detecting the Correct Answer

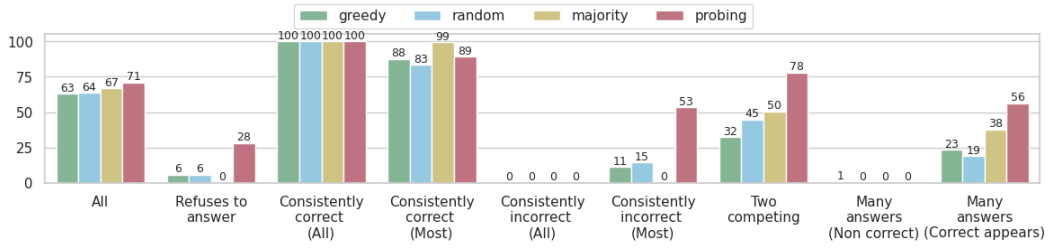
After identifying that models encode diverse truthfulness-related information, we examine how this *internal* truthfulness aligns with their *external* behavior during response generation. To this end, we use our probe,[5] trained on error detection, to select an answer from a pool of 30 generated responses to the same question. We then measure the model's accuracy based on the selected answers. A case where this accuracy does not significantly differ from traditional decoding methods (such as greedy decoding), suggests that the LLM's internal representation of truthfulness is consistent with its external behavior. In simpler terms, that the model is generating answers that it also internally considers as correct. Conversely, a case where using the probe alters performance either way, would suggest a misalignment between the LLM's internal representations and its actual behavior.

Experimental Setup The experiments were conducted on TriviaQA, Winobias, and Math. We resample each model answer in the same strategy described in Section 5.1. The final chosen answer is the one with the highest correctness probability, as assessed by the probe. We compare to three baselines: (1) greedy decoding, (2) random selection from the $K = 30$ answer candidates; and (3) majority vote wherein the most frequently generated answer is chosen.

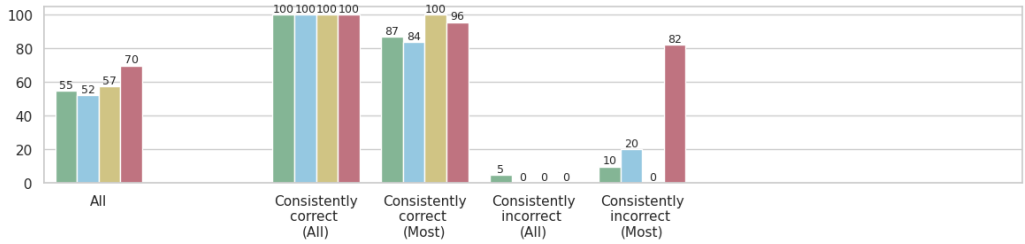
Results The results for Mistral-7b-instruct are summarized in Figure 5, with additional results for other LLMs and datasets as well as qualitative examples provided in Appendix E. We only present results on error types that appear 30 times or more in our test dataset. Overall, using the probe to select answers enhances the LLMs accuracy across all examined tasks. However, the extent of improvement varies by error type. For instance, in the TriviaQA dataset, there is minimal gain in the "mostly correct" category (B2). In contrast, substantial gains—ranging from 30 to 40 points in some cases—are observed in the "mostly incorrect" (C2), "two competing answers" (D), and "many answers" (E1) categories. Interestingly, and perhaps surprisingly, the probe is most effective in cases where the LLM lacks any (external) preference for the correct answer during generation. The fact that the probe can effectively identify the correct answer in these scenarios, points at a significant disconnect between the LLM's internal encoding and its external behavior. These results suggest that even when the model encodes information of which answer is correct, it can still generate an incorrect answer in practice.

While using the probe to select the answer proves effective, it is not proposed here as an error mitigation strategy but rather as a diagnostic tool. However, these findings indicate that further research in this area could leverage the existing knowledge within LLMs to significantly reduce errors. We recommend exploring this direction in future investigations.

[5] We choose the best-performing probe for each task, which is trained on the last exact answer token.



(a) TriviaQA



(b) Math

Figure 5: Different answer choice strategies, Mistral-7B-Instruct. A notable improvement in accuracy by using the error-detection probe is observed for error types where the LLM shows no preference for the correct answer across repeated generations.

7 Discussion and Conclusions

In this study, we analyzed LLM errors through their internal representations. Our approach depends on access to internal representations, restricting its use to open-source models. We focus on QA tasks with clear gold labels, which are key for benchmarking truthfulness detection and valued by the community. To ensure robustness, we tested 10 datasets across 4 model architectures. Open-ended tasks are left for future research, with our work laying the groundwork for broader applications. For instance, we found that truthfulness-related information is localized in specific tokens within long-responses, enabling practical improvements in error detection for production models. This insight could extend to tasks like summarization, by probing the most meaningful entities in an answer.

Truthfulness features showed poor generalization across tasks and datasets, highlighting the need for caution when applying trained error detectors in varied settings. Some unexplained patterns suggest hidden links between unrelated tasks that warrant further research. Improving generalization could involve exploring the effects of layer-token combinations and training on diverse datasets, as demonstrated by Bürger et al. (2024). Deciphering task-specific truthfulness features and their overlaps across tasks might also enhance classifier design. Still, task-specific probes could be highly valuable in critical fields like medicine and law, where reliability matters. These probes can detect errors, predict error types, and guide response selection from resampled outputs, offering significant practical benefits. Guidelines for applying these probes are provided in Appendix F.

Finally, we identified a significant discrepancy between the model's external behavior and internal states, where it repeatedly outputs incorrect responses despite internally encoding the correct answer. It is possible that mechanisms favoring likelihood override those promoting truthfulness, as LLMs are trained to predicting likely tokens, which does not necessarily align with factual accuracy. Our findings imply that these models already encode valuable information that could possibly be harnessed to reduce errors. Work by Chuang et al. (2024) shows promising results in this area, while a subsequent work by Gekhman et al. (2025) focused exclusively on this "hidden knowledge" phenomenon, formally defining it and studying its extent. In conclusion, our findings suggest that LLMs' internal representations provide useful insights into their errors, highlights the complex link between the internal processes of models and their external outputs, and hopefully paves the way for further improvements in error detection and mitigation.

8 REPRODUCIBILITY STATEMENT

To ensure reproducibility of our work, we provide detailed instructions and necessary code. The source code, including scripts for generating model answers, probing, resampling, and error type analysis, is available in the supplementary material, where we also provide command examples and specific seeds used for experiment reproducibility. This repository includes documentation on how to set up the environment, download and preprocess datasets, and execute the experiments outlined in Sections 3–6 of the paper. Additionally, all datasets, models, and results generation steps are described in the Appendix A.

ACKNOWLEDGMENTS

This research was supported by the Israel Science Foundation (grant No. 448/20), an Azrieli Foundation Early Career Faculty Fellowship, an AI Alignment grant from Open Philanthropy, and a Google gift. HO is supported by the Apple AIML PhD fellowship. This research was funded by the European Union (ERC, Control-LM, 101165402). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

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